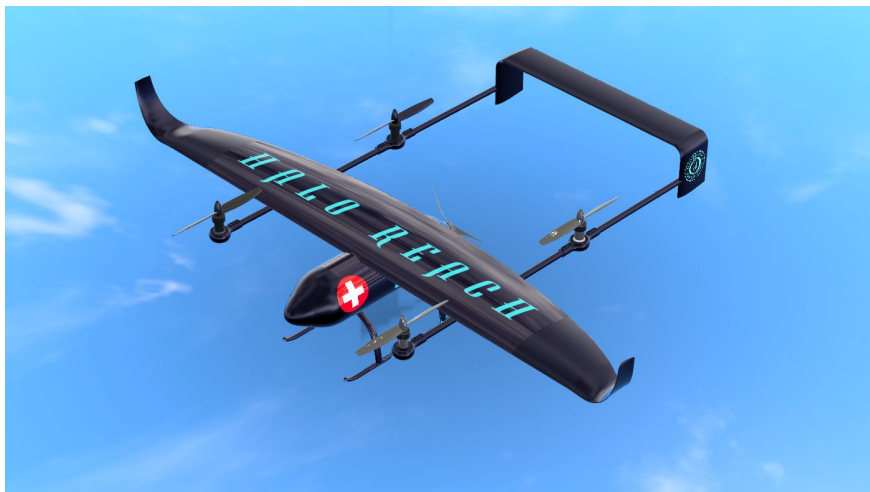

HALO REACH: Launch and Recovery System Design and Actuator Design

UAS for Disaster Relief - AE3 2024



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Abstract

As a member of system design team, this report is mainly focus on designing the launch and recovery system and actuator system. The launch and recovery system includes a motion compensation system using Stewart platform. Its range of motion can achieve 60, 74 and 83 degree in roll pitch yaw range of angle. In addition, a $\pm 1.5m$, $\pm 1.5m$ and $\pm 3.4m$ range of motion of surge sway and heaves can be achieved. This gives a reliable stable platform for landing. The positioning and vision based landing technology further improve the landing successful rate, reach up to 97.42 %. Micro controller was selected with high PWM channels of 24 and high CPU speed of 200MHz. In addition, its low energy consumption rate of 1.2ma make it to be optimal choice. Servo mechanism is developed for three diferent control surface. By proper selecting the linear motor and calculate the geometry of design.It finally can satisfy the requirement of up to 30 Nm for elevators, 10 Nm for Aileron and 15 Nm for Rudder.

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Nomenclature

$\dot{\theta}$	Deflection Angular Velocity of Control Surface(Degree/s)
$\dot{x}$	Speed of Stroke (m/s)
θ	Deflection Angle of Control surface(Degree)
F	Maximum Force produced by Linear motor(N)
F_i	Force act on Component e (N)
T	Maximum Torque of Control Surface (Nm)
a	Length of Component a (mm)
b	Length of Component b (mm)
b	Length of Component c (mm)
d	Length of Component d (mm)
e	Length of Component e (mm)
GTOW	Gross Takeoff Weight
HALO REACH	Highly Aerodynamic Light Operation Remote Emergency Aid & Crisis Help
LARS	Launch and Recovery System
UAS	Uncrewed Aerial System
UAV	Unmanned Aerial Vehicle
VTOL	Vertical Take-Off and Landing
x	Length of Stroke (m)

1 Introduction

1.1 Requirements and Constraints

The report investigates and proposes the design of the launch and recovery system (LARS) for a ground control system used in an unmanned aircraft system (UAS), as well as the actuator design. This system is intended for disaster relief operations at sea. The ocean environment and the nature of disaster relief missions require more considerations than typical UAS operations, such as supply delivery. The mission profile specifies that the launch and recovery footprint should be six by six meters, and the UAV should be launched and recovered vertically from the deck of a ship. The UAV should be operable by only two people, and modular add-ons should be swapped out in less than 10 minutes [1].

The requirements stated in the mission profile are insufficient to design a LARS due to the harsh environment. Firstly, a ship experiences six degrees of freedom motion due to strong ocean waves, including roll, pitch, yaw, surge, heave, and sway [2]. Numerous studies have attempted to predict ship motion to improve the performance of motion compensation systems [3]. Companies such as STABLE have developed motion compensation products for drone landing platforms on ships [4]. Therefore, motion compensation is essential for safe drone landings without collisions with the ship's deck.

Secondly, strong gusts on the ship can occur due to varying weather conditions. Safety regulations state that vessels should not be operated if gusts exceed 55 miles per hour [5]. Wind speeds during hurricanes can reach up to 72 miles per hour. Therefore, a mechanism to secure the UAV on the launch platform is necessary to prevent collisions caused by the UAV being shifted by gusts.

Thirdly, seawater and storms can damage the UAV on the deck. This is a common occurrence [6]. Thus, a waterproof design for the UAV's protective environment is necessary, and anti-corrosive materials should be chosen to improve the system's lifespan.

1.2 Current State of the Art

Before discussing the design, it is important to investigate the current state of the art. [7] discusses the control design for a rope-hook mechanism. When the UAV collides with a rope set up on the vessel, its velocity decreases, and it hangs on the rope. The advantages of this mechanism are its low risk, especially in challenging weather conditions, and its ability to handle high landing speeds due to the rope minimizing the UAV's speed. The required equipment is easy to set up and small in size. However, this design is not suitable for our mission. Firstly, it is only applicable for light UAVs, whereas our UAV has a maximum takeoff weight (MTOW) of 160 kg. Secondly, it only addresses recovery, meaning an additional launching system is still needed.

[8] discusses various types of UAV launch and recovery systems for vessels. For takeoff, UAVs can be launched using bungee cords, hydraulic launchers, pneumatic launchers, or rocket-assisted take-off. Figure 3a shows a typical bungee catapult. These methods require a rail and eject the UAV like a catapult, preventing vertical launches. Although commonly used in vessel operations, these methods require shifting a 160 kg UAV to the launcher, which may require more than two people and significant time if payload swapping is needed under time constraints.



(a) Bungee Catapult



(b) Net Recovery

Figure 3. Common Launching and Recovery Systems

For recovery, UAVs can be captured using nets, Skyhook, or windsocks. These techniques are simple to manufacture

and similar to the rope-hook method. However, they can easily damage the UAV structure and are difficult to set up. Additionally, UAVs need to be removed from the net, wasting time and requiring more people. Figure 3b shows a typical net recovery system, which is commonly used for recovering UAVs from vessels.

In general, comparing different types of launch and recovery systems against the mission requirements and operating context indicates that an innovative LARS design is needed to meet all the requirements and constraints. The current state of the art is insufficient for a disaster relief mission.

2 Launch and Recovery System Architecture

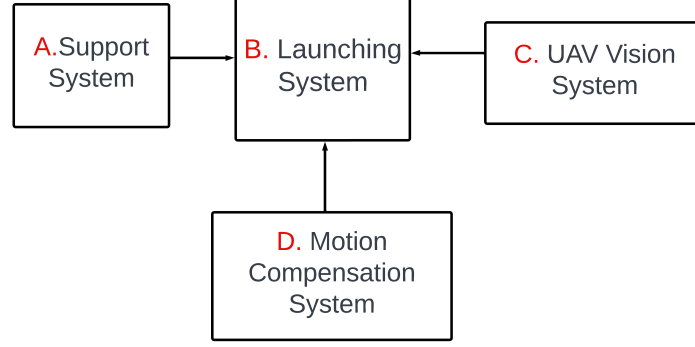


Figure 4. Overall System Architecture of LARS

The Figure 4 demonstrates the overall architecture of LARS. Each block represents a detailed subsystem. This section briefly introduces the function of each block and explains how it addresses the requirements outlined in the introduction section. As shown in Figure 4, the system is divided into four subsystems: A. Support subsystem, B. Launching subsystem, C. UAV vision subsystem, and D. Motion compensation subsystem. The UAV vision subsystem is on-board, while the other three are built on the ground.

In A, the support system mainly includes a waterproof hangar used to store the UAV while it waits to reach the take-off spot. The system also contains a small vehicle used for transporting the UAV to the launch and recovery platform. This vehicle also has a function to swap the cubic-shaped payload on the UAV. In B, the launching system mainly includes an anti-impact and anti-corrosive material platform. Additionally, a ski-type positioning mechanism is included to correct the position of the landing UAV. The fixer on the platform is designed to secure the ski-type landing gear UAV, preventing it from being shifted by gusts. There is also a heating system that can form an 'H' pattern on the platform, assisting landing and ensuring safety in low visibility environments, such as at night or in bad weather.

In C, the on-board system mainly includes the IR and standard cameras. As mentioned previously, the standard camera is used to detect the pattern on the platform to determine the heading direction and height. The IR camera is used to detect the heat pattern in low visibility contexts. In D, the motion compensation system is a 6 DOF motion system that compensates for the motion of the ship due to waves and gusts. This system contains sensors, a controller, and a Stewart platform capable of 6 DOF motion. The sensors are mainly inertial measurement units, including accelerometers, magnetometers, and gyroscopes. These sensors measure the state of the ship and transmit the data to the controller, which adjusts the platform to maintain a steady, fixed horizontal position.

3 System Design

3.1 Motion Compensation System

Figure 7a illustrates the architecture of the motion compensation system. IMU sensors on the ship transmit data to the controller. After sensor fusion, the controller uses the state and inverse kinematics of the Stewart Platform, along with the desired state of the platform, to calculate the desired stroke length. Sensors on the platform measure its current state and transmit this data to the UAV via a telemetry module on the ship to assist UAV flight control. The state is also sent to the controller. The controller calculates the actual stroke length based on the current state and adjusts the platform using the error between the desired and actual stroke lengths.

Figure 5a shows the design of the Stewart platform used for launch and recovery. The platform can perform six degrees of freedom (6 DOF) motion and is also widely used in aircraft simulators. [9] discusses the dynamics and kinematics of the Stewart platform based on its geometry. Due to its flexibility, it is an ideal choice for a vessel motion compensation system. The letters A, B, C, and D are labeled for detailed explanation. The joints at A and D are universal joints with 2 DOF flexibility. B and C are linear actuators with a stroke. The stroke length of each leg defines the platform's 6 DOF motion. The linear actuators can be either hydraulic or electric motors.



Figure 5. Stewart Platform Drawing

The sizing of each component can be optimized further. The dynamic and kinematic performance is correlated to the design, but optimizing the sizing within the page limit is not feasible. The current sizing of each component is provisional and used to analyze the provisional range of motion of each 6 DOF. Figure 6 shows the rough sizing of each component. Figure 6a shows the side-view of the Stewart platform, and Figure 6b shows the top-view. Using these rough sizings, the range of motion of the 6 DOF can be evaluated.

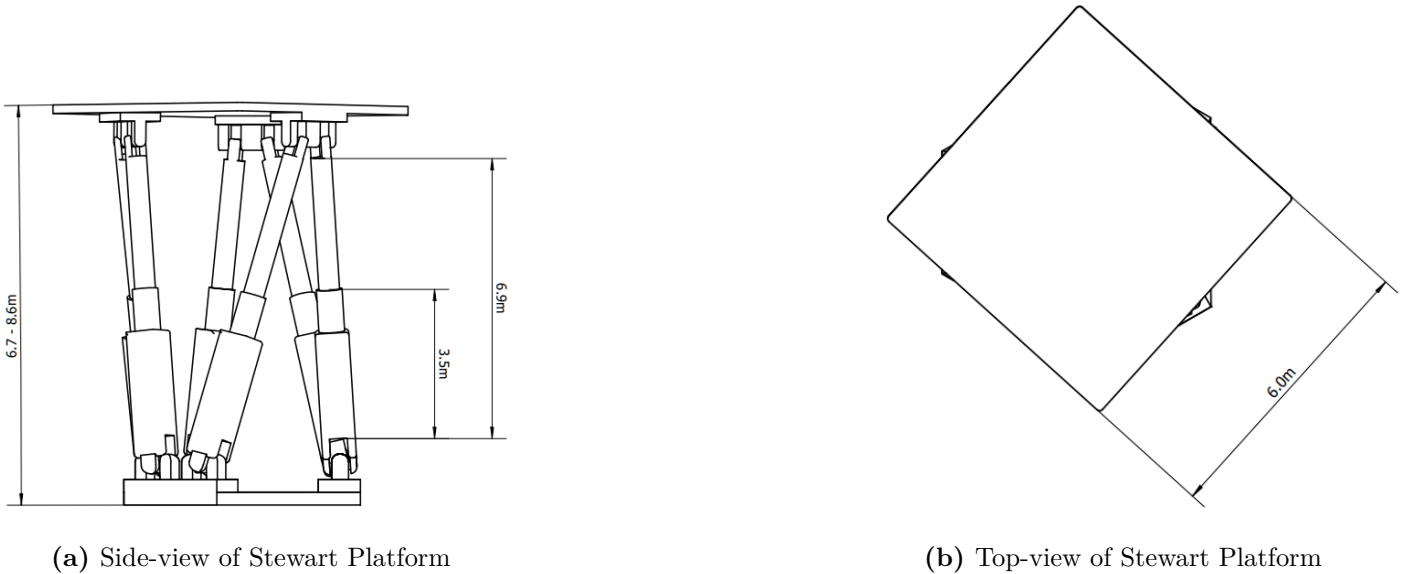


Figure 6. Rough Sizing of Stewart Platform

Table 3 compares the range of motion of the Stewart platform derived from its geometry with the range of motion of a ship [10]. Note that the ship's motion is due to wave disturbances. The weather conditions are severe but assumed feasible for operations, excluding hurricanes. Table 2 provides data on a product developed by Motion System [11].

According to the tables, the range of motion of our Stewart platform meets the requirements perfectly. However, compared to the commercial product, our Stewart platform is much larger. Although the commercial product's range of motion is slightly smaller than required, it can carry a payload of 800 kg, which is five times the weight of

Table 2. Specifications for the PS-6TL-800

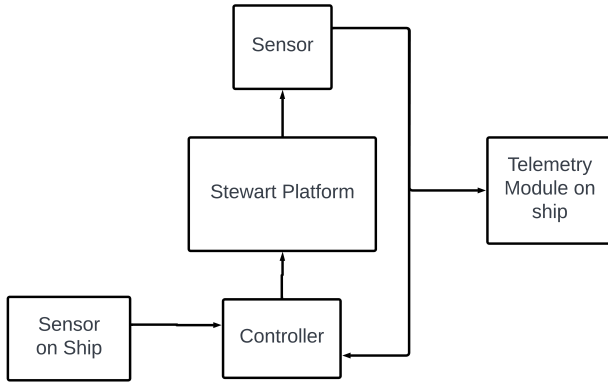
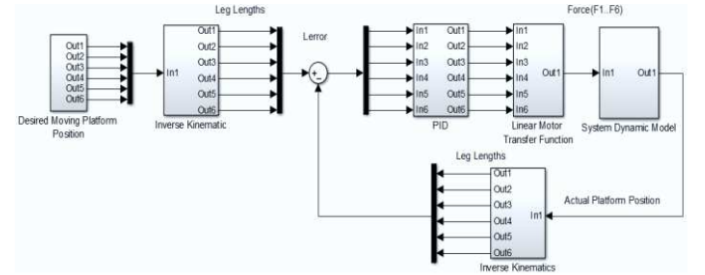
Moving Load (kg)	Payload (kg)	Length (m)	Width (m)	Height (m)
800	720	2.12	1.90	1.07

Table 3. Motion Data for Different Platforms

	Surge (m)	Sway (m)	Heave (m)	Roll (deg)	Pitch (deg)	Yaw (deg)
Stewart	-1.5 to 1.5	-1.4 to 1.5	-3.4 to 3.4	-31 to 47	-34 to 39	-45 to 38
Ship Motion	-0.58 to 0.89	-0.78 to 0.63	-0.46 to 0.53	-27 to 30	-25 to 25	-10 to 10
PS-6TL-800	-0.37 to 0.35	-0.33 to 0.33	-0.18 to 0.22	-24.5 to 24.5	-23.6 to 21.8	-37.0 to 37.0

the UAV. Therefore, further optimization of the Stewart platform's size, range of motion, torque requirements, and payload capacity is necessary. Note that the PS-6TL-800 is not specifically designed for UAV launch and recovery systems and costs approximately 51,500 Euros.

[9] presents a typical PID control of a Stewart platform based on its dynamic and inverse kinematic model. The paper starts by deriving a dynamic model based on the geometry of the Stewart platform using vectors and rotational matrices. After determining the leg lengths based on the relationship between the platform and the base, the research derives the inverse velocity analysis to find the velocity and angular velocity of the legs. Subsequently, a comprehensive dynamic analysis is carried out using Lagrange formulation. Finally, based on the dynamics and kinematics, a PID control design is implemented. Figure 7b shows the control system using MATLAB Simulink. The system performs accurately with a low settling time and minimal error. When using a linear actuator, high-frequency oscillations are observed, but the system eventually stabilizes. These effects are due to the non-linearity of the mechanism dynamics. Therefore, proper gain selection based on the characteristics of the linear motor and system is necessary. Verification of this system is complex and could occupy extensive pages. Due to time and page limitations, using this reference for verification of its dynamics and control system is the most appropriate approach.

**(a)** System Architecture of Motion Compensation System**(b)** System block diagram with linear motors.**Figure 7.** System architecture and Control system of Stewart platform

3.2 Launching System and UAV Vision System

In this section, the Launching system and UAV vision system are researched. The architecture is shown in Figure 8. The launching system contains launching pad, fixer, positioning mechanism, heating system and platform pattern. The launching pad should be an anti-impact and anti-corrosive material so can robust enough when experiencing the impact by UAV and preventing saline and alkaline environment corrosion. The positioning mechanism is being used to correcting the landing position of UAV. The fixer can grab and fix the Ski-type landing gear on the platform and make sure the UAV is fixed when swapping payload or charging and fueling on a rocking vessel. The heating system and pattern have the same purpose. Heating system is aim to emit the infrared ray of a certain pattern while the pattern reflect the visible light. The on-board system includes IR and standard camera, an image processing program, and a program transform image to heading direction. The IR and standard camera capture the infrared ray pattern and visible light pattern from launching pad. Infrared ray is mostly used in a low visibility

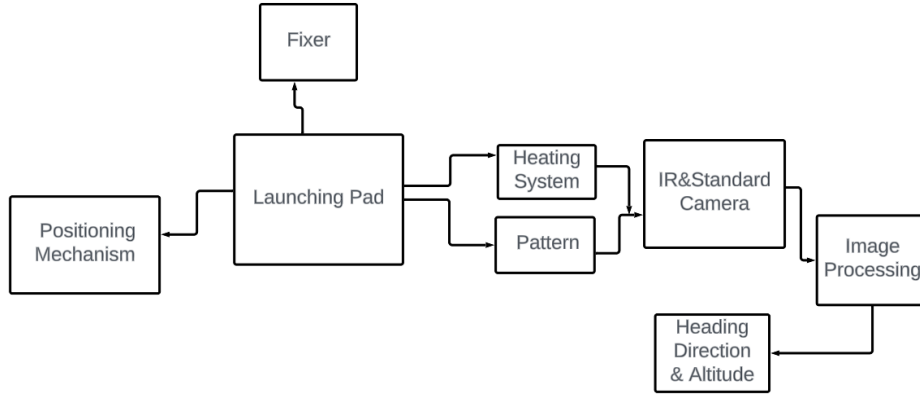


Figure 8. Architecture of Launching and Vision System

condition such as bad weather or at night. The image processing program transform the image captured into a black-white color image. Then another program using this image compare with the template pattern and find out the differences. Therefore the heading direction can be obtained from pattern orientation differences. The altitude can be obtained from pattern's size differences.

3.2.1 Positioning Mechanism

There are numerous drone landing platforms on the market equipped with positioning mechanisms. This is due to several reasons. Firstly, accurate landing positioning enables automatic wireless charging of the drone. Secondly, it facilitates information exchange between the UAV and the computer on the landing platform. Thirdly, precise positioning is essential for safe and efficient loading and unloading of payloads. Therefore, positioning mechanisms are crucial for meeting these requirements.

In our mission profile, a positioning mechanism is necessary to help the UAV land accurately during gusty weather, allowing the fixer to secure the UAV on the platform, ensuring a safe operational environment. The following section analyzes current positioning mechanisms in the market and designs a suitable mechanism for our mission profile.

Figure 9 demonstrates four commonly used types of positioning mechanisms. An active positioning mechanism involves moving parts like motors or actuators to correct the position, while a passive mechanism relies on the platform's geometry, such as a contour or funnel, to correct the drone's position. The design of the mechanism depends entirely on the shape of the landing gear. Figures 9a and 9c present active positioning mechanisms. After the drone lands on the platform, the pusher moves the UAV to an accurate position in one or two directions. The V-shape pusher pushes in one direction while the V shape corrects the position in another direction. Figures 9b and 9d show passive mechanisms. The mechanism in Figure 9b is designed for four-legged drones, with a concave platform shape that guides the legs to the center of the funnel. The mechanism in Figure 9d is designed for ski-type legs, using fixed sloped surfaces to correct the parallel ski-legs to the middle area of the platform.

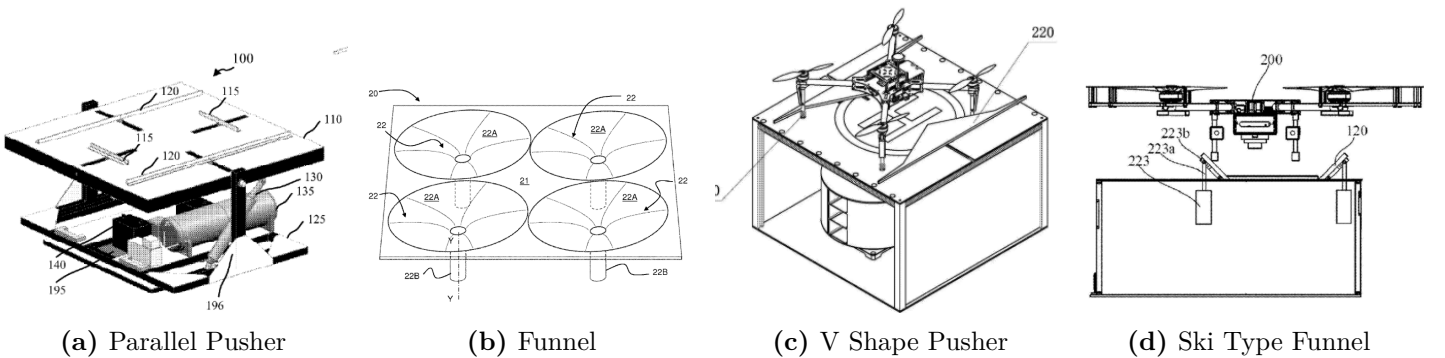


Figure 9. Different Types of Pushers and Funnels

Regarding the current UAV design, the UAV weighs a maximum of 160 kg and has a ski-type landing gear configuration. Considering the UAV's weight, a pusher may not be applicable due to high friction. Even if the pusher has sufficient power, the ski-type landing gear could be worn down. Therefore, a ski-type funnel should be used to

match the UAV's design.

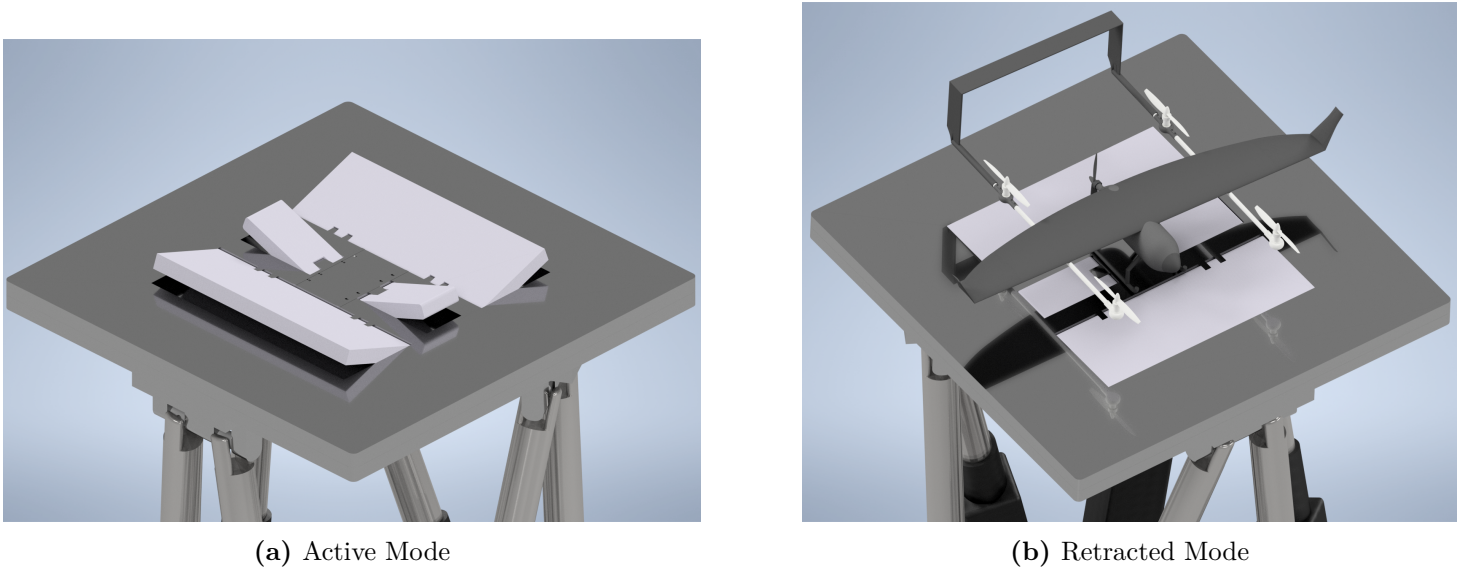


Figure 10. The Active Ski-Type Leg Positioning Mechanism with Demonstration of UAV

Combining designs from references and our mission profile requirements, an innovative positioning mechanism is shown in Figure 10. The slope can be activated to form a 20-degree incline, positioning the UAV during landing in two directions. The slope is driven by a servo motor. After landing, the slope retracts back to the platform, allowing operational space. The slope is only activated during strong gusts and significant ship motion. If the flight control and ship motion compensation systems can ensure landing accuracy in good weather, the landing platform can remain in retracted mode. This customizable function allows the platform to operate in various contexts, including fast-changing ocean weather. The slope is sized to accommodate a landing error of up to 1.2 meters in both directions. The maximum impact force on the platform is defined by the maximum relative momentum between the slope and the UAV, and this force is used to select a suitable servo motor.

This positioning mechanism lacks complete feasibility data, requiring ship motion data, optimized compensation system data, and UAV turbulent landing speed data (considering gusty and unsteady flow). However, its large error tolerance and reasonable design based on research and requirements suggest that this positioning mechanism is partially reliable and usable for disaster relief missions.

3.2.2 Visible and IR Pattern and Vision Based Autonomous Landing System

The above sections discussed how hardware assists in the safe landing of UAVs. This section introduces software technology. [12] explored various types of vision-based autonomous landing in different contexts, such as runways, moving platforms (vehicles or ships), and more complex scenes. Vehicle platform movement can be simplified to two dimensions, but on the complex and dynamic sea, ships exhibit three-dimensional movement due to environmental factors. This requires real-time resolution of six degrees of freedom for the relative position between the UAV and the ship, demanding high speed and accuracy from the UAV system. Additionally, sea surface mist and weather significantly affect the imaging quality of visual sensors, complicating autonomous landing on carrier-based platforms.

Historical Development [12]

- **2003:** Qiu conducted autonomous landing of UAVs using a binocular stereo vision system, initially meeting practical requirements but struggling in maritime environments.
- **2009:** Xu introduced a heating module on the "T" logo to mitigate water mist effects, allowing infrared cameras to achieve a 17.2 ms recognition speed and 97.2% accuracy for night landings [13].
- **2014:** Sanchez-Lopez utilized Hu invariant moment features and decision tree discriminant units to detect international landing signs, achieving reliable target recognition under varied conditions [14].
- **2015:** Portuguese researchers employed airborne RGB cameras and Kalman filters to estimate the UAV's position, velocity, and attitude, generating the flight trajectory for landing. This method faced challenges due to water reflections [15].

- **2017:** Polvara proposed a scheme combining extended Kalman filtering and inertial navigation for accurate pose estimation, but it was not tested in real environments. Subsequent simulated experiments showed robustness in adverse weather conditions [16].
- **2020:** Li et al. proposed a new feature extraction structure using SSD and KCF algorithms with deep learning, achieving a detection success rate of 91.1% and an average processing speed of 9 ms, suitable for UAV autonomous landing [17].

Current Focus and Challenges

Researchers are primarily focused on addressing the three-dimensional motion of moving platforms and improving cooperative target recognition. Despite significant progress in detection accuracy and real-time performance, current vision algorithms struggle to resolve relative poses without relying on cooperative targets. Extending algorithms to detect the entire platform could enhance detection distance, adaptability, and robustness of vision systems for UAV autonomous landing. The table in [12] lists performance precision, processing speed, height, and image resolution, comparing different markers and algorithms. Due to space constraints, the table can be found in the reference. By comparing different technologies, a platform design was selected to enable landing.

Landing Procedure

The algorithm based on [18] involves the following steps:

1. Real-time capture of the marker or information from the landing area.
2. Process the image to perform feature point calculation, feature extraction, feature matching, and other operations to achieve landmark tracking and detection.
3. Calculate the attitude and height from the difference between the real-time pose data on the UAV and the marker pose data.
4. Use the calculation for flight control to adjust the UAV's landing speed and attitude in real-time.
5. Successfully land on the platform.

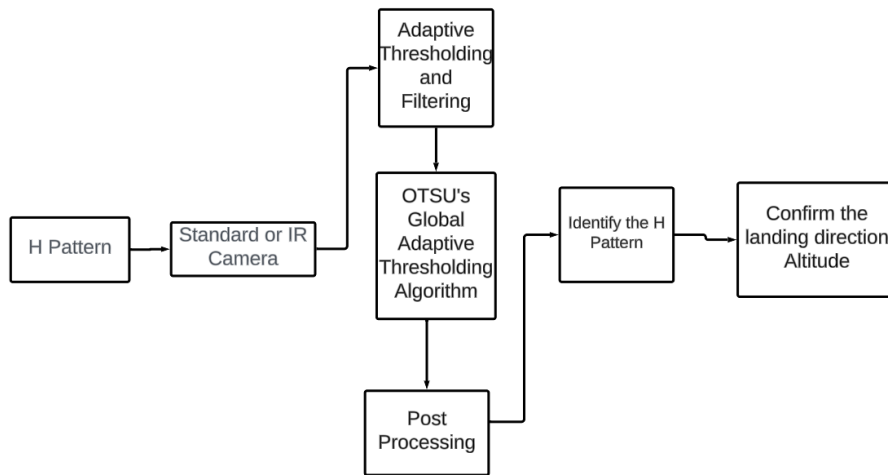


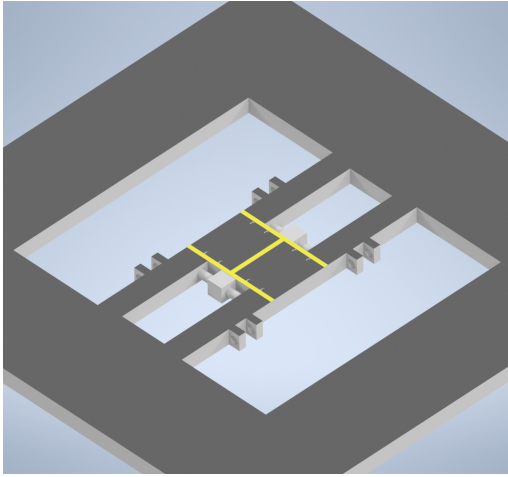
Figure 11. Basic vision based processing procedure was selected based on research

The image processing algorithm shown in Figure 11 should transform the original image from Figure 12a into Figure 12b. The algorithm first captures an image from a camera or IR camera and converts it into a grayscale image. The next step is to apply the Otsu algorithm, which lists all grayscale pixels and classifies them into two classes: the target set and the background class. This stage uses within-class variance and between-class variance iteratively to calculate the threshold. Finally, it turns the image into a binary image. Note that the image obtained from the IR camera is a heat map, with the heating wire forming an H-shaped pattern. The IR camera is expected to provide better landing performance than a visible light camera, as it can be used in low visibility environments such as fog or night. Additionally, it can also be used in sunny weather because reflective sunlight from the ocean can interfere with the visible light camera.

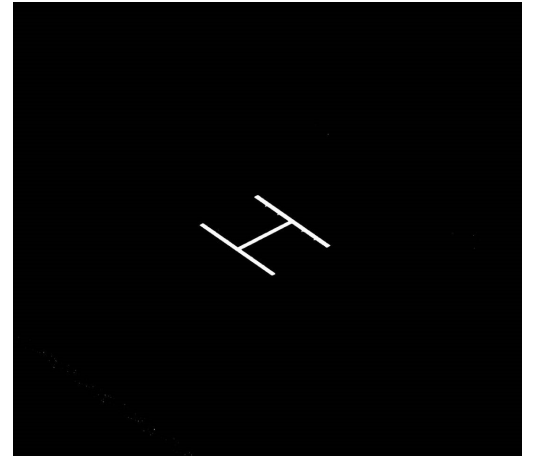
According to [18], the next step is to identify several corner points of the H pattern and transform them into coordinates. By calculating the coordinate differences between points on the same edge, the slope can be determined, and the rectangle can be fitted. Finally, the fitted rectangle image is rotated and compared to the template H pattern in memory. If the error is below a threshold value, the direction of heading can be found. Furthermore, zooming in on the picture can help estimate the altitude. The research presents the results of this method in the following tables. The successful detection rate is 97.42%. However, 24% of the time is consumed in processing the image. Additionally, because the image is processed on a ground control station, a significant amount of time is wasted in transmission. Therefore, future plans include improving the algorithm and making it accessible on-board the UAV.

Table 4. Errors of target recognition and Computational cost 15 FPS image processing[18]

Errors of target recognition algorithm		Computational cost 15 FPS image processing	
Total Number of Frames	15662	Image Processing	CPU time
Landpad present	8205	Image Acquisition	$\approx 20\%$
Landpad detected	7993	Adaptive Thresholding and Filtering	$\approx 16\%$
		Image Segmentation	$\approx 10\%$
		Image Registration	$\approx 24\%$
		GUI Interface & data display	$\approx 30\%$



(a) Original Image



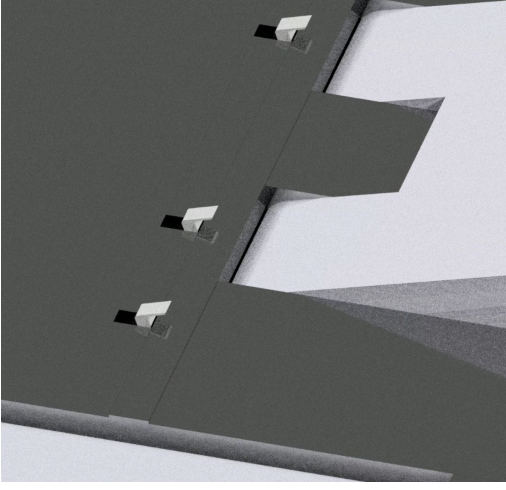
(b) Filtered Image

Figure 12. The desired result of filtering algorithm

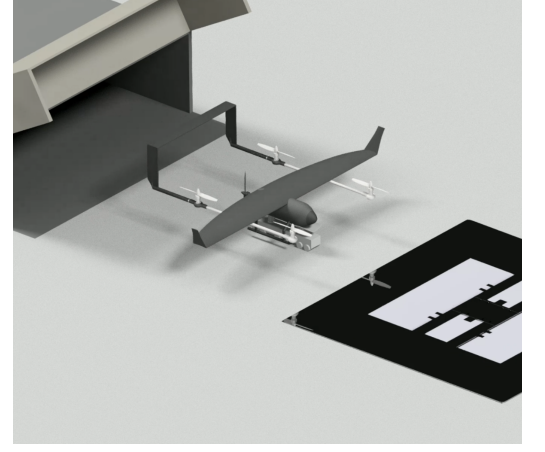
3.3 Future Improvement and Plan

A fixer, shown in Figure 13a, is a crucial component that secures the UAV on the vessel. It protects the UAV from disturbances caused by ship motion and strong gusts. The following figure illustrates the basic concept of the fixer. After the UAV lands on the platform using the positioning mechanism, eight clamps extend to secure the ski-type landing gear. The motion compensation system can then be stopped, allowing for charging and refueling. This technology might only be used in adverse weather conditions or large waves. In the future, the motor used in the fixer should be selected based on the torque generated by the UAV due to the strongest gusts. Additionally, structural and material analysis should be conducted to optimize performance and size. A system capable of automatically deploying the fixer with the positioning mechanism should be developed. A non-slip material should be selected for the contact surface between the landing gear and the fixer. Further analysis is required to verify its feasibility.

Figure 13b also illustrates the design of a hangar and a transport vehicle, which is still conceptual. The goal is to store the UAV and protect it from seawater on the deck. The storage room, 2 meters high, allows for safe maintenance and fueling. The transport vehicle can lift the UAV from the bottom of the fuselage and transfer it to the launch and recovery platform. In future plans, both the hangar and vehicle require detailed design, including the power system of the vehicle, waterproof mechanisms, and material selection for anti-corrosive purposes.



(a) Initial Fixer Design



(b) Support System Conceptual Design

Figure 13. Initial Design of Fixer and Support System

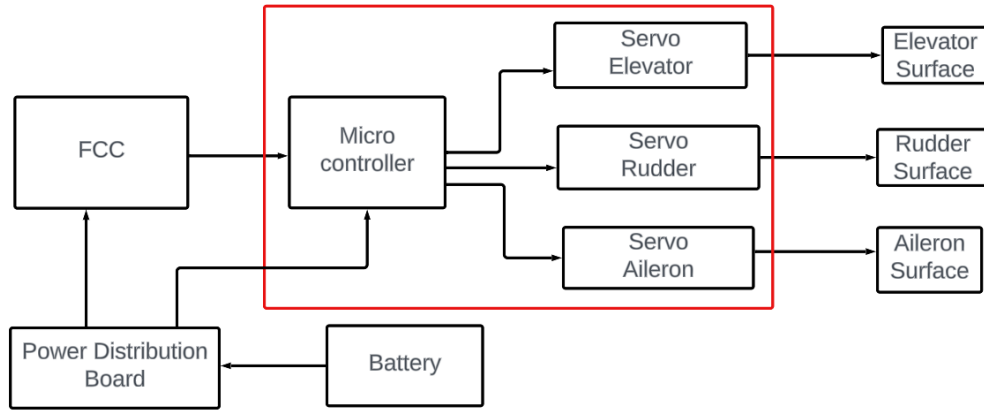


Figure 14. Actuator System Architecture (In Red Box)

4 Actuator Design

This section investigates, selects, and designs the actuator system. The architecture of the actuator system is shown in the red box in Figure 14 below. Outside the red box is a simplified architecture of the UAV system. The accurate avionics system should be based on [19]. Inside the red box, the actuator is divided into two parts: the selection of the microcontroller and the design of the servo. To select a microcontroller for UAV actuator control, it is essential to understand the requirements: identify the actuator types (rudder for yaw, elevator for pitch, and aileron for roll) and ensure the microcontroller has sufficient PWM outputs and ADC inputs for control and feedback. Evaluate the processing power needed for control algorithms and determine necessary communication protocols (e.g., CAN, I2C, SPI, UART). Ensure the microcontroller meets aerospace standards like DO-178C and DO-254, and verify compatibility with development tools and support from the manufacturer. This ensures the microcontroller can handle real-time processing and communication needs effectively. The servo design includes selecting a suitable servo motor and developing a mechanism that meets the following requirements: torque, range of angle, and angular velocity. This ensures good control performance.

4.1 Microcontroller Selection

The specifications for selecting a microcontroller include the number of PWM channels, ADC channels, CPU speed, memory, operating temperature, supply voltage, cost, and power consumption (PC). Table 5 compares three types of microcontrollers from three series: STM32F4 [20], Texas Instruments [21], and PIC32MZ Series [22]. After comparing the STM32, TMS32, and PIC32MZ microcontrollers, the TMS32 stands out as the optimal choice. The TMS32 chip has the highest CPU speed, the highest number of PWM channels, and the lowest energy consumption. Its RAM memory is also the highest, although its flash memory is 1 MB lower than the PIC32. More importantly, the TMS32 is designed for real-time control. The highest number of channels allows the controller great flexibility in controlling multiple actuators simultaneously. Although its price is approximately 25 pounds,

its superior real-time performance, design specifically for control applications, and compliance with DO-178C and DO-254 aerospace standards make it a more robust and reliable choice. However, considering potential weaknesses such as development complexity, it is prudent to select an additional microcontroller for redundancy. The STM32, robust performance, and compliance with aerospace standards, serves as an excellent secondary option. This dual-microcontroller approach ensures operational safety and reliability for our UAV actuator control system, even in the face of unforeseen challenges. This approach provides a balanced perspective, acknowledging the strengths of the TMS32 while also addressing potential concerns and ensuring system redundancy with the STM32.

Table 5. Microcontroller Options for UAV Actuator Control

Specification	STM32F407VG	TMS320F28379D	PIC32MZ2048EFH144
Size	14.2 mm x 14.2 mm x 1.4 mm	17.2 mm x 17.2 mm	20 mm x 20 mm
Supply Voltage	1.8V to 3.6V	3.3V	2.3V to 3.6V
PWM Channels	Up to 12	24	16
CPU Speed	168 MHz (ARM Cortex-M4)	200 MHz (C28x DSP core)	200 MHz (MIPS core)
Memory	192 KB RAM, 1 MB Flash	1 MB RAM, 1 MB Flash	512 KB RAM, 2 MB Flash
Real-time Performance	Excellent with DSP and FPU	Designed for real-time control	High-performance with extensive support
Energy Consumption	1.7 mA/MHz	1.2 mA/MHz	2.0 mA/MHz
Price	~10.00 GBP	~25.00 GBP	~12.00 GBP

4.2 Servo Design

Servos in the actuator system are essential for controlling the control surfaces, such as the aileron, elevator, and rudder. Each of these control surfaces contributes to the stability of the entire UAV. Therefore, it is crucial to consider all control stability factors when designing the servo. In a typical aircraft, control surfaces are usually operated using hydraulic systems or steel wires. However, in UAVs, a servo motor or a linear motor with some mechanical transmission mechanism is used. To propose a feasible design, specifications such as torque, range of motion, and angular velocity are required. The specifications are shown in Table 6. The aileron requires a torque of 7 Nm with a range of angle deflection of ± 25 degrees. The elevator requires a torque of 25 Nm and a range of angle deflection of ± 30 degrees. The rudder requires a torque of 13 Nm and a range of angle deflection of ± 25 degrees. All of them should achieve a deflection from full negative to full positive within one second. Therefore, an angular change rate of 60 degrees per second is set as the specification.

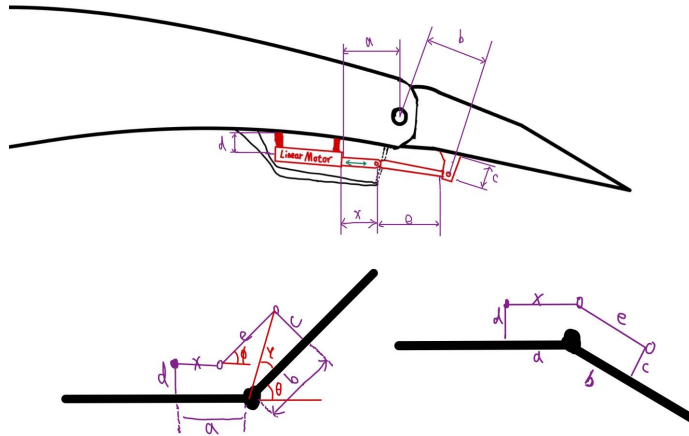


Figure 15. Control Surface Mechanism and Its Geometry

The geometry is shown in Figure 15. The geometry is simplified to demonstrate the geometric relationship. x is the stroke length of the linear actuator, and it is variable. The following three formulas are derived based on

the geometry. They describe the relationship between the deflection angle and stroke length, stroke velocity and deflection angular velocity, and torque and stroke length. The proper length of each component and type of linear actuator should satisfy the above requirements and the formulas. Additionally, the length is constrained by the chord length of each control surface.

$$e^2 = (a + b \cos \theta - c \sin \theta - x)^2 + (-d + b \sin \theta + c \cos \theta)^2 \quad (1)$$

$$0 = 2(a + b \cos \theta - c \sin \theta - x)(-\dot{\theta}b \sin \theta - \dot{\theta}c \cos \theta - \dot{x}) + 2(-d + b \sin \theta + c \cos \theta)(\dot{\theta}b \cos \theta - \dot{\theta}c \sin \theta) \quad (2)$$

$$T = F_i e \cos \left(90 - \left(\theta + \tan^{-1} \left(\frac{c}{b} \right) \right) - \tan^{-1} \frac{(-d + b \sin \theta + c \cos \theta)}{(a + b \cos \theta - c \sin \theta - x)} \right) \quad (3)$$

where

$$F_i = F \cos(\tan^{-1} \frac{(-d + b \sin \theta + c \cos \theta)}{(a + b \cos \theta - c \sin \theta - x)})$$

a, b, c, d, e are the lengths of each component

x is the stroke length

F is the maximum force produced by the linear actuator

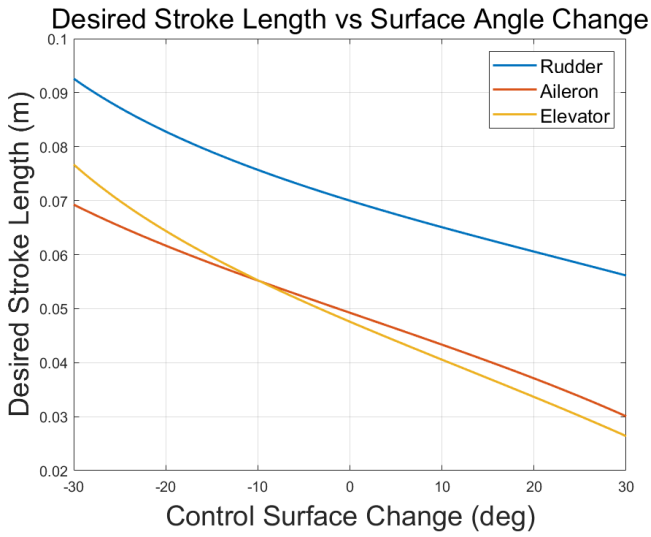
F_i is the maximum force experienced on the e bar

The following procedure is followed: first, estimate a reasonable length of different components based on the chord length constraint. Secondly, using equations 1 and 2, input the range of angle and desired angular velocity to find the range of stroke length and stroke velocity. Then, using equation 3, substitute the calculated values and the desired torque to determine the desired force distribution. Figure 16 presents the required stroke length, velocity, and force distribution along the deflection angle change from -30 to 30 degrees. The code used for the calculation is in Appendix B. Next, search online and note down the linear actuators that satisfy the range, velocity, and force requirements. Note that the size of the linear actuator also needs to be considered. Finally, select the type of linear actuator that meets the requirements. Appendix A lists the parameters of three linear motors. If none of them are feasible, redo the entire procedure from the first step. Note that the elevator can be operated by two servos, allowing for a smaller linear actuator. Table 6 below presents the results for each of the three control surfaces. Pay attention: the angular speed and torque are the minimum torque generated throughout the deflection process, as they change with different deflection angles. Therefore, a control system that maintains a steady angular velocity is required. Moreover, the last row of the table shows how many servos are required to control one surface.

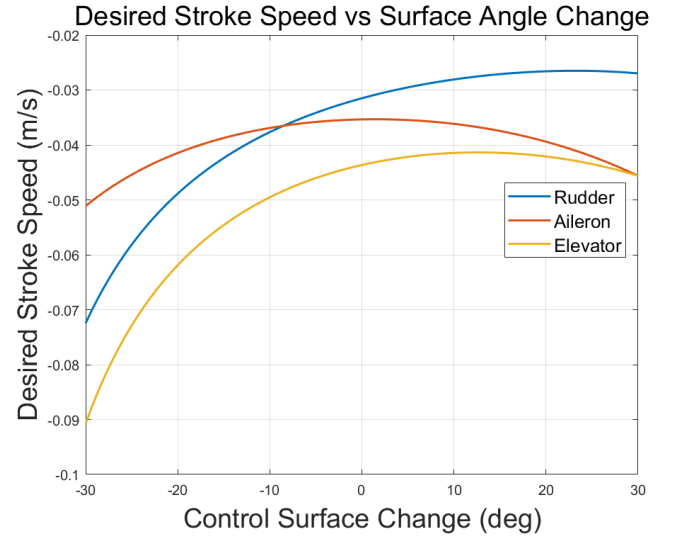
5 Conclusion

This report discusses several important designs for the launch and recovery system and actuator system. The Stewart platform has its trusted landing error tolerance because of the length of stroke can provide high range of motion in 6 Dof. However, the size can be optimized, the real response time and stability for our mission still required a further research and develop. The proper selection in linear motor and a energy save and space save solution still could be improved in the future. The active ski-type positioning mechanism is designed based on the actual requirement of our mission and refer to passive ski-type mechanism. The concept is trustworthy and reliable for our mission as this mechanism is proven by research. However, a more detail design and further experimental verification should be carried out to prove its robustness in a gusty condition. The motor choosing, deflection angle are some parameters can be considered further. Vision based autonomous is also a proven technology that give a great help in landing, it can successfully identify the pattern and calculate the parameter it needs. But still, the algorithm can be still further optimized to achieve a faster response time and higher accuracy identifying the pattern in low visibility.

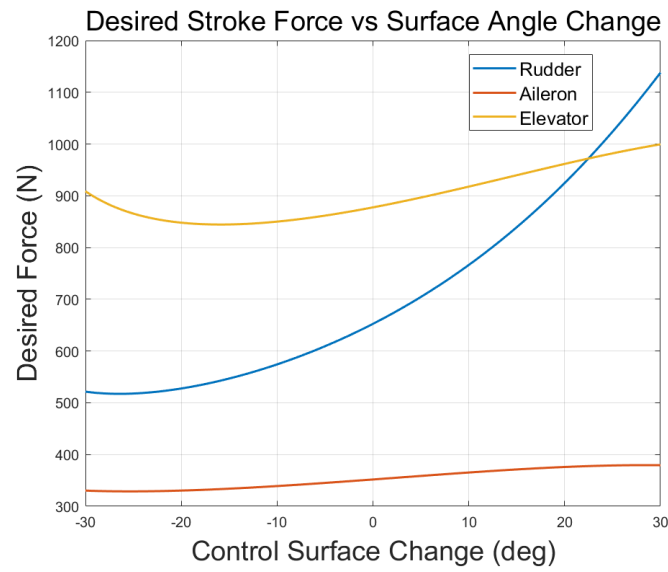
In the actuator section, a micro-controller has been roughly selected, based on several important feature like channel number, CPU speed, memory. Those standard ensure the micro-controller's functionality and satisfy the basic requirement. Next step is considering its compatibility with other system and its robustness in extreme condition. For the servo system, based on the geometry and dynamic, the linear actuator that make whole system satisfied the requirement is selected. In addition, the mechanical structure is also roughly developed. Next stage is probably further develop the CAD and its control system. As can be seen in the graph, a steady rate of changing angle an stable torque required different force and speed in different deflection angle. Moreover, a proper optimization of component length might also achieve better performance and lower energy consumption.



(a) Desired Stroke Length



(b) Desired Stroke Speed



(c) Desired Stroke Force

Figure 16. Desired Parameters for Choosing Linear Actuator

Table 6. Performance and Specification of Rudder, Aileron, and Elevator

	Performance			Specification		
	Rudder	Aileron	Elevator	Rudder	Aileron	Elevator
Torque (Nm)	15	10	30	13	7	25
angle range (deg)	± 30	± 30	± 30	± 25	± 25	± 30
angle rate (deg/s)	60	60	60	60	60	60
Motor Type	TiMOTION TA19	LINAK LA36	LINAK LA20	N/A	N/A	N/A
a(mm)	50	45	34	N/A	N/A	N/A
b(mm)	100	80	100	N/A	N/A	N/A
c(mm)	30	40	30	N/A	N/A	N/A
d(mm)	30	34	30	N/A	N/A	N/A
e(mm)	80	76	87	N/A	N/A	N/A
Number of Servo	2	2	2	N/A	N/A	N/A

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Appendix

A Selection of Linear Motor

Table 7. Comparison of Linear Actuators

Actuator Model	Force (N)	Stroke Length (cm)	Voltage (V)	Energy Consumption (W)	Cost (USD)
TiMOTION TA19 [23]	1100	9	12/24	30	150
LINAK LA36 [24]	1000	8	12/24/36/48	45	200
LINAK LA20 [25]	350	7	12/24	20	100

B Matlab Code for Calculating Desire Parmeter of Linear Motor

```

1 a=[0.05 0.045 0.034];
2 b=[0.1 0.08 0.1];
3 c=[0.03 0.04 0.03];
4 d=[0.03 0.034 0.04];
5 e=[0.08 0.076 0.087];
6 the=linspace(-30,30,100);
7 T=[15 10 30];
8 for j=1:3
9     syms x
10    for i=1:100
11
12    eqn=(a(j)+b(j)*cosd(the(i))-c(j)*sind(the(i))-x).^2+(-d(j)+b(j)*sind(the(i))+c(j)*cosd(the(i))).^2-e(j)^2==0;
13
14    sol([1:2],i) = solve(eqn,x);
15
16
17    end
18    syms xd
19    for i=1:100
20        eqn1=2*(a(j)+b(j)*cosd(the(i))-c(j)*sind(the(i))-sol(1,i))*(-1.05*b(j)*sind(the(i))-1.05*c(j)*cosd(the(i))-xd)+2*(-d(j)+b(j)*sind(the(i))+c(j)*cosd(the(i)))*(1.05*(b(j)*cosd(the(i))-c(j)*sind(the(i))))==0;
21    sol2(i)=solve(eqn1,xd);
22    end
23
24    syms f
25    for i=1:100
26        eqn2=f*cosd(atan2((-d(j)+b(j)*sind(the(i))+c(j)*cosd(the(i)))/(a(j)+b(j)*cosd(the(i))-c(j)*sind(the(i))-sol(1,i)))))*e(j)*cosd(90-((the(i)+atan2(c(j)/b(j))-atan2((-d(j)+b(j)*sind(the(i))+c(j)*cosd(the(i)))/(a(j)+b(j)*cosd(the(i))-c(j)*sind(the(i))-sol(1,i))))))-T(j)==0;
27        sol3(i)=solve(eqn2,f);
28    end
29    so(j,:)=double(sol(1,:));
30    so2(j,:)=double(sol2);
31    so3(j,:)=double(sol3);
32
33
34
35    end
36    figure(1)
37    for j=1:3
38        plot(the,so(j,:), 'LineWidth', 1.5);
39    hold on
40    end

```

```

41 hold off
42 title('Desired Stroke Length vs Surface Angle Change','FontSize', 19);
43 xlabel('Control Surface Change (deg)','FontSize', 19);
44 ylabel('Desired Stroke Length (m)','FontSize', 19);
45 legend('Rudder', 'Aileron', 'Elevator','FontSize', 13);
46 grid on;
47
48 figure(2)
49 for j=1:3
50 plot(the,so2(j,:), 'LineWidth', 1.5);
51 hold on
52 end
53 hold off
54 title('Desired Stroke Speed vs Surface Angle Change','FontSize', 19);
55 xlabel('Control Surface Change (deg)','FontSize', 19);
56 ylabel('Desired Stroke Speed (m/s)','FontSize', 19);
57 legend('Rudder', 'Aileron', 'Elevator','FontSize', 13);
58 grid on;
59
60 figure(3)
61 for j=1:3
62 plot(the,so3(j,:), 'LineWidth', 1.5);
63 hold on
64 end
65 hold off
66 title('Desired Stroke Force vs Surface Angle Change','FontSize', 19);
67 xlabel('Control Surface Change (deg)','FontSize', 19);
68 ylabel('Desired Force (N)','FontSize', 19);
69 legend('Rudder', 'Aileron', 'Elevator','FontSize', 13);
70 grid on;

```

Listing 1. Desired Stroke Length vs Surface Angle Change